GLASSBOX: PERSONALIZED CALIBRATION FOR PRIVACY-PRESERVING, EXPLAINABLE ONLINE EXAM PROCTORING

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Abstract*—Online examinations have become central to education, and cheating attempts have grown alongside this shift. Existing browser- and camera-based proctoring tools typically judge every student against one fixed, population-wide behavioral standard. This has caused documented unfairness: facial recognition components flag darker-skinned students more often due to training-data bias, and disabled or neurodivergent students are flagged for normal behavior such as looking away or needing to step away. This makes fair, privacy-respecting, and explainable exam monitoring a pressing problem. We propose GlassBox, a browser-based add-on proctoring layer that combines MediaPipe Face Landmarker for real-time gaze and head-pose tracking with tab-switch, focus-loss, fullscreen-exit, and copy-paste signals. Rather than judging students against a shared population norm, each student is calibrated against their own individual baseline, captured through a guided 20–30 second sequence at the start of the exam; later readings are scored as deviation from that personal baseline rather than a fixed global threshold, and accommodated students receive individually shifted thresholds instead of a separate rule. Raw video and audio never leave the browser — only small derived evidence values are sent onward. These evidence signals are grouped into temporal episodes and fused into features that feed an interpretable Logistic Regression classifier, trained on a self-recorded, scenario-labeled dataset, producing a transparent Low/Medium/High risk band with a per-signal explanation for the instructor. As a separate, offline fairness audit — not part of the live pipeline — we compare detection-confidence disparity between MediaPipe and the open-source RetinaFace detector on the FairFace dataset to check whether skin-tone bias appears consistently across independently trained models. This extended version additionally details the model structures used, the governing formulas of the calibration and scoring pipeline, a system demo walkthrough, and a structured comparative analysis against commercial and academic baselines. Compared to commercial systems such as Proctorio and Examity, GlassBox's contribution is the combination of personalized calibration, privacy preservation, and measured, explainable fairness evaluated together as one system. A key limitation is that, since GlassBox relies on browser-based signals rather than continuous video surveillance, a student using an undetected second device outside the camera frame would not trigger any signal, regardless of instructor attentiveness.

Index Terms—Online exam proctoring, gaze tracking, personalized calibration, algorithmic fairness, explainable AI, privacy-preserving monitoring, logistic regression, model interpretability.

I. INTRODUCTION

The shift to online and remote examinations has made automated proctoring tools common, but most existing tools apply a single fixed detection standard uniformly to every student. Peer-reviewed and journalistic reporting has documented two recurring harms of this approach: facial recognition components trained predominantly on lighter-skinned faces produce lower confidence and more false flags for darker-skinned students, and behavior-based flags penalize disabled or neurodivergent students for actions — such as looking away, stimming, or needing to step away — that are normal for them but deviate from a population norm. Tools that additionally record continuous video and audio raise separate privacy concerns; a 2022 U.S. court ruling found mandatory webcam room-scans unconstitutional, and data-protection frameworks such as India's DPDPA treat this kind of biometric data as requiring extra care.

This paper presents GlassBox, a browser-based, privacy-preserving, and explainable risk-scoring add-on layer for online exam proctoring, designed as a college mini-project. GlassBox does not replace a full exam platform; it is a layer that can sit alongside one, in the same spirit as commercial tools such as Proctorio, but built around personalized calibration rather than a fixed population threshold.

Beyond the original proposal, this extended report adds four things that a purely conceptual write-up cannot convey on its own: (i) the internal structure of every model in the pipeline, so a reader can judge what is learned versus what is fixed; (ii) the exact formulas used to turn raw signals into a calibrated risk band, so the scoring logic is auditable rather than a black box; (iii) a walkthrough of the working demo as currently implemented; and (iv) a structured, criterion-by-criterion comparison against the systems introduced in Section II, rather than a purely narrative one.

II. RELATED WORK

Yaqub, Mohanty, and Suleiman [1] proposed a privacy-preserving eye-gaze tracking approach for online proctoring that calibrates a personalized model per examinee to improve gaze-tracking accuracy. GlassBox is positioned as an extension of this idea: instead of using per-student calibration only to improve tracking accuracy, GlassBox repurposes the same calibration step as a behavioral-fairness

baseline, adds an explainable scoring layer on top, and evaluates the result on a purpose-built dataset.

AutoOEP [2] uses an LSTM over multiple signals to reach 90.7% detection accuracy, but relies on continuous dual-camera video and identity verification, which is not privacy-preserving. A related U.S. patent [3] claims baseline-calibrated proctoring in general terms but is not peer-reviewed and reports no accuracy figures or fairness evaluation. The commercial system Proctor360 markets "individual behavioral baselines" but publishes no technical methodology for independent evaluation. Established commercial platforms such as Proctorio and Examity already provide percentage- or color-coded risk scores, but neither performs personalized per-student calibration, preserves privacy by keeping raw video off the server, nor provides signal-level explainability.

Positioning

GlassBox's contribution is not a new detection algorithm — Logistic Regression is a standard, well-understood classifier. The contribution is (i) using deviation from a personal, guided-calibration baseline as the input representation instead of a raw or population-normalized signal; (ii) deliberately choosing an interpretable model over a more complex one because proctoring decisions are high-stakes and must be appealable; and (iii) evaluating privacy preservation, personalized calibration, and explainability together as one system, which no prior tool or paper in this section does jointly. Section VII places this positioning into a structured comparison table.

III. PROPOSED SYSTEM ARCHITECTURE

Fig. 1 shows the full GlassBox pipeline. The architecture is split into a client stage, which never transmits raw biometric data, and a server stage, which converts small derived signals into a calibrated, explainable risk band. Each stage is described below, including the formulas that govern the transformation from raw signal to final decision.

A. Client-Side Signal Collection

A Browser Signal Collector captures tab-switch/visibility-change events, fullscreen-exit, and copy-paste attempts (blocked outright via preventDefault, not scored), while a Vision/Gaze Collector runs MediaPipe Face Landmarker — a pretrained, unmodified Google model — to extract 478 facial landmarks, including iris position, at each frame, together with a per-frame confidence value. Only small derived evidence values are sent to the server — raw video and audio never leave the browser.

B. Guided Calibration and Personal Baseline

At the start of the exam, a guided 20–30 second sequence (look center, left, right, up, down) is used to compute a personal baseline for gaze vector g and head-pose (yaw, pitch, roll) per student. For each channel x, the mean μ_x and standard deviation σ_x are computed over the calibration window of N frames:

$$\mu_x = (1/N)\,\Sigma\, x_i\,, \quad \sigma_x = \sqrt{[(1/N)\,\Sigma\,(x_i - \mu_x)^2]} \qquad (1)$$

Every subsequent frame is then converted into a personal deviation score rather than a raw value, using a z-score-style normalization:

$$d(t) = |\, x(t) - \mu_x \,| \,/\, (\sigma_x + \varepsilon) \qquad (2)$$

where $\varepsilon > 0$ is a small constant that prevents division by zero for channels with near-zero calibration variance. Because d(t) is expressed in units of the student's own variability, a naturally fidgety student and a naturally still student are held to the same relative standard rather than the same absolute threshold — this is the core fairness mechanism of GlassBox.

C. Server-Side Evidence Engine and Temporal Episodes

Raw signals are never treated as cheating evidence on their own; they pass through an Evidence Engine and are grouped into Temporal Episodes using a 1.5-second gap threshold, so that closely spaced events are interpreted as one behavioral episode rather than independent flags. Formally, two consecutive evidence events at times t_i and t_{i+1} belong to the same episode if:

$$t_{i+1} - t_i \le 1.5\ s \qquad (3)$$

and a new episode begins otherwise. Each episode e is summarized by its duration, peak deviation score, and the set of signal types it contains.

D. Fusion, Feature Engineering, and Model

A Fusion Engine and Feature Engine combine episodes across signal types into a model-ready feature vector $x = [x_1, \ldots, x_k]$, including mean gaze deviation, head-pose deviation, episode frequency per minute, mean episode duration, tab-switch count, and fullscreen-exit count. These features feed a Logistic Regression classifier, chosen for interpretability. The model computes a linear score and passes it through the logistic (sigmoid) link function:

$$z = w \cdot x + b = \Sigma\, w_i\, x_i + b \qquad (4)$$

$$p = \sigma(z) = 1 \,/\, (1 + e^{-z}) \qquad (5)$$

where $p \in (0, 1)$ is the model's estimated cheating-risk probability for the session. Training minimizes the standard binary cross-entropy loss over the labeled episode dataset (Section V):

$$L(w, b) = -(1/M)\,\Sigma\,[\, y_j \log p_j + (1 - y_j) \log(1 - p_j)\,] + (\lambda/2)\,\|w\|_2^2 \qquad (6)$$

The added $(\lambda/2)\|w\|_2^2$ term is an L2 penalty on w, included directly in the training objective to discourage over-reliance on any single signal and improve generalization on the small self-recorded dataset; $\lambda \ge 0$ is the regularization strength.

E. Risk Banding and Accommodation Shifts

The Risk Engine converts p into a Low/Medium/High band — never a raw numeric score — using two thresholds τ_1 and τ_2 (defaults 0.35 and 0.70, as shown in Fig. 5a):

$$band(p) = Low \text{ if } p < \tau_1;\ Medium \text{ if } \tau_1 \leq p < \tau_2;\ High \text{ if } p \geq \tau_2 \quad (7)$$

Accommodated students receive individually shifted thresholds using the same formula rather than a separate rule: $\tau_1' = \tau_1 + \Delta$ and $\tau_2' = \tau_2 + \Delta$, where Δ is a small, disclosed, instructor-approved offset. This keeps the scoring logic identical for every student — only the operating point moves.

F. Explainability Layer

Because Logistic Regression is linear in its inputs, each feature's contribution to the score is directly readable as $c_i = w_i\ x_i$, and the XAI Explanation layer reports the signals with the largest $|c_i|$ as the "reason" shown to the instructor — for example, "gaze deviation contributed 61% of this session's score." All sessions, evidence, episodes, and risk scores persist in a SQLite database, and only Medium/High sessions are surfaced to a human reviewer on the Teacher Dashboard; no exam is ever auto-terminated by the system.

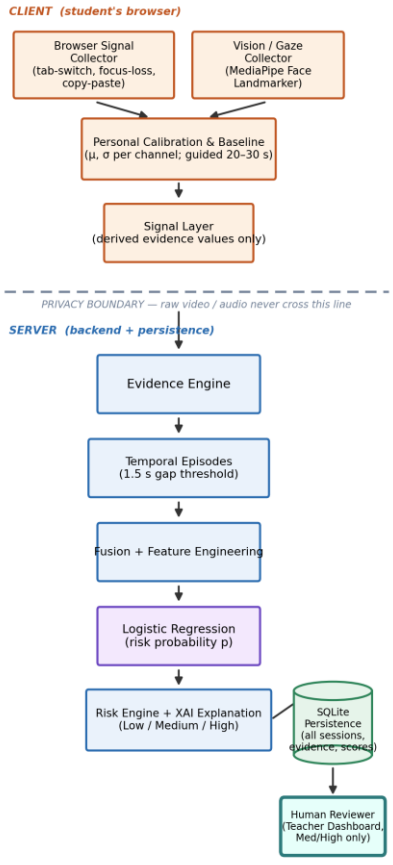


Fig. 1. GlassBox end-to-end architecture, redrawn to make the client/server split, the calibration stage, and the privacy boundary explicit. Color coding is kept consistent across all figures in this paper: orange = client-side, blue = server-side processing, purple = trained ML model, green = persistent storage, teal = human reviewer.

IV. MODEL STRUCTURES AND MODELS USED

GlassBox does not train any new detector from scratch. It composes three existing models with well-documented internal structures, summarized in Table I and Fig. 4, and adds one small model it trains itself — the Logistic Regression risk classifier.

A. MediaPipe Face Landmarker (live pipeline)

MediaPipe Face Landmarker is used unmodified, at inference time only. Internally it chains a lightweight BlazeFace-style detector (an SSD-style convolutional network that regresses face bounding boxes at real-time speed) with a landmark-regression CNN that predicts 478 3-D facial landmarks, including a dense iris sub-mesh, from the cropped face region. Head-pose (yaw/pitch/roll) is then derived geometrically from stable landmark triplets, and gaze direction from the iris landmarks relative to the eye contour, as shown in Fig. 4(a). GlassBox uses only the numeric landmark and confidence outputs — the frame itself is discarded immediately after inference.

B. RetinaFace (offline fairness audit only)

RetinaFace is used only in the separate, offline fairness audit of Section V, never in the live proctoring path. It uses a ResNet-50 or MobileNet-0.25 backbone feature extractor, a Feature Pyramid Network that fuses features across scales P2–P5, and a shared context module feeding multi-task heads that jointly predict a face-classification score, bounding box, 5-point landmarks, and dense 3-D vertex regression, shown in Fig. 4(b). Because RetinaFace is trained independently of MediaPipe on different data, agreement between the two on the same fairness pattern is stronger evidence than either model alone.

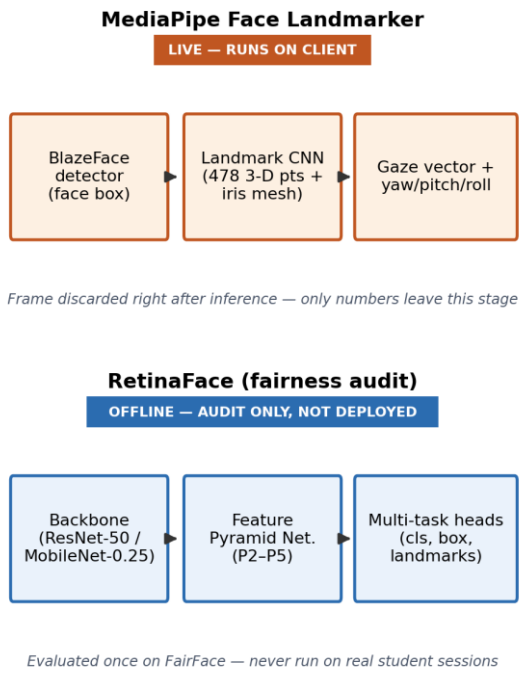


Fig. 4. Simplified internal structure of the two pretrained vision models used by GlassBox, using the same color language as Fig. 1 — orange marks the live, client-run MediaPipe pipeline; blue marks the offline, audit-only RetinaFace pipeline.

C. Logistic Regression Risk Classifier (trained by the team)

The only model GlassBox trains itself is the risk classifier from Section III-D. Structurally it is a single linear layer of size k (the feature count) followed by a sigmoid activation — no hidden layers — which is precisely why it is chosen: every weight w_i is directly interpretable as the influence of one behavioral feature. Fig. 5(a) shows this as a minimal interpretability flow — fused features, a weighted sum, a sigmoid, and a risk band — deliberately kept free of dense equation clutter, and Fig. 5(b) shows an illustrative decision surface over two fused features, with the shaded regions corresponding to the Low/Medium/High bands.

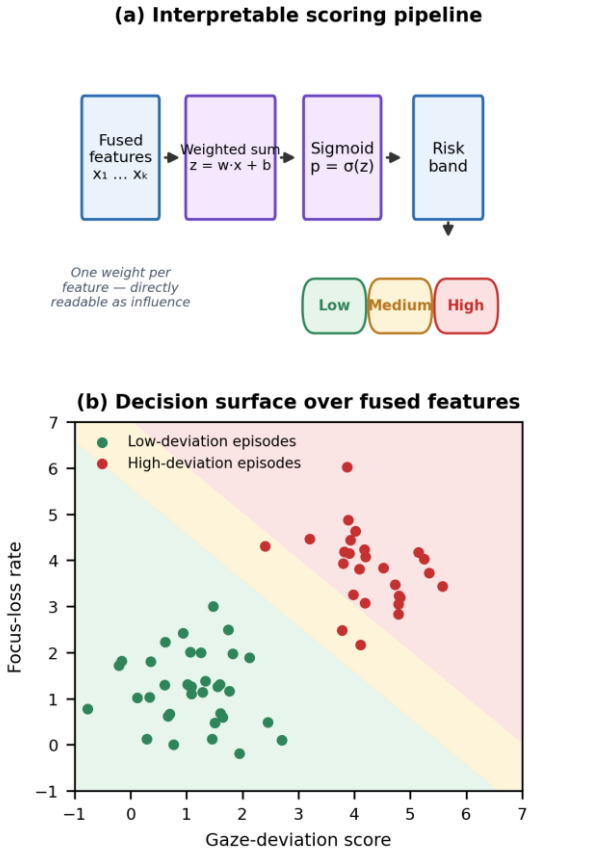


Fig. 5. Logistic Regression structure — (a) a minimal features → weighted-sum → sigmoid → risk-band flow highlighting per-feature interpretability, and (b) an illustrative decision surface over two fused features.

TABLE I
MODELS USED ACROSS THE GLASSBOX PIPELINE

Model	Role in GlassBox	Trained?	Structure
MediaPipe Face Landmarker	Live gaze/head-pose extraction	Pretrained, frozen	BlazeFace + landmark CNN
RetinaFace	Offline fairness audit only	Pretrained, frozen	Backbone + FPN + multi-task heads
Logistic Regression	Risk scoring (Low/Med/High)	Trained by team	Linear layer + sigmoid

V. DATASET

Because MSU OEP is video-only and cannot represent tab-switch, fullscreen, or copy-paste signals — and would undermine the explainability goal — GlassBox trains on a self-recorded dataset generated with the application itself: three team members each ran five scripted sessions (Normal, Looking-away, Head-turn, Tab-switch, Combined), giving approximately 15 sessions and an estimated 150–200 evidence episodes. Sessions are labeled at the session level via a manifest (session_id, person, scenario, label), avoiding manual frame-by-frame labeling; each teammate's local SQLite database is exported to CSV and merged before training. No synthetic data is used.

The FairFace dataset is used separately, only for an offline fairness audit: MediaPipe and RetinaFace (used here

purely as a pretrained, unmodified evaluation tool, not fine-tuned and never deployed on real student sessions) are each run once on FairFace to compare detection-confidence disparity across skin tones, checking whether the same bias pattern appears in two independently trained models. Table II summarizes the two datasets and their distinct purposes.

TABLE II
DATASETS USED AND THEIR PURPOSE

Dataset	Used for	Size
Self-recorded (GlassBox)	Training/eval of Logistic Regression	~15 sessions, ~150–200 episodes
FairFace [4]	Offline detector fairness audit	Balanced by race, gender, age

VI. EVALUATION PLAN

Evaluation covers four separate tests. (1) Software testing: unit and integration tests of each signal and the end-to-end flow. (2) ML accuracy: Precision, Recall, F1, and ROC-AUC against a fixed-threshold baseline, using k-fold cross-validation given the small sample size, with both false-positive and false-negative rates reported. (3) Fairness audit: the MediaPipe/RetinaFace confidence disparity on FairFace, reported honestly rather than claimed as solved. (4) Human reviewer trial: approximately five reviewers assess a mixed set of sessions, including deliberately wrong or borderline flags, measuring both review speed and whether reviewers catch mistakes, as a check for automation bias.

The accuracy metrics in (2) are computed in the usual way from true/false positives and negatives (TP, FP, TN, FN):

$$Precision = TP / (TP + FP), \quad Recall = TP / (TP + FN) \tag{8}$$

$$F1 = 2 \times (Precision \times Recall) / (Precision + Recall) \tag{9}$$

ROC-AUC is computed as the area under the True-Positive-Rate vs. False-Positive-Rate curve swept over all decision thresholds, giving a threshold-independent measure of separability between genuine and deviating episodes.

Fig. 6 shows the intended shape of the two key evaluation figures using illustrative, non-measured values so the reporting format can be reviewed before real numbers exist; both panels will be regenerated from actual results before final submission, as flagged in their captions.

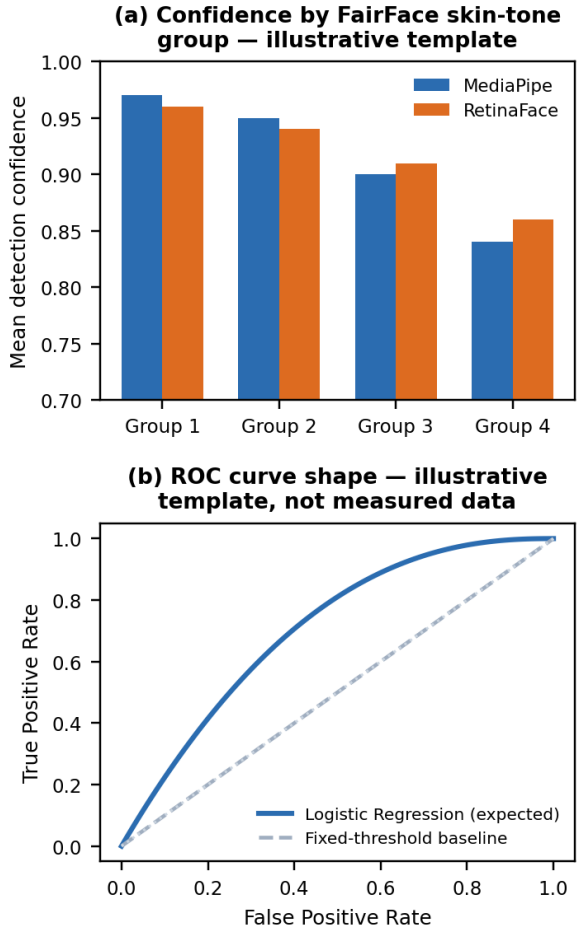


Fig. 6. Illustrative evaluation-figure templates, NOT measured results — (a) expected shape of the MediaPipe vs. RetinaFace detection-confidence comparison across FairFace skin-tone groups, and (b) expected ROC-curve shape for the Logistic Regression model against the fixed-threshold baseline. Both will be replaced with real k-fold results before submission.

VII. COMPARATIVE ANALYSIS

Table III turns the narrative positioning of Section II into a direct, criterion-by-criterion comparison. A system earns a check only if the capability is documented and independently verifiable, not merely marketed.

TABLE III
CRITERION-BY-CRITERION COMPARISON

System	Personal calib.	Privacy	Explain.	Fairness audit
GlassBox	✓	✓	✓	✓
Proctorio	—	—	—	—
Examity	—	—	—	—
AutoOEP [2]	—	—	—	—
Proctor360	✓	—	—	—
Yaqub et al. [1]	✓	—	—	—

(Privacy = raw video never leaves the device; Explain. = signal-level explainability). GlassBox is the only system satisfying all four columns; Proctor360 and Yaqub et al. [1] satisfy only personalized calibration.

Read narrowly, Table III does not claim GlassBox is more accurate than AutoOEP's reported 90.7% — that comparison is deferred to Section VI once GlassBox's own

k-fold numbers exist. It claims only that GlassBox is the sole system evaluated jointly on calibration, privacy, and explainability, which is the specific gap identified in Section II.

VIII. DEMO AND SYSTEM WALKTHROUGH

The current implementation is a React + Vite frontend with MediaPipe Tasks Vision integrated, plus a backend that persists sessions to SQLite. A student's flow is: (1) join the exam link, (2) complete the 20–30 second guided calibration sequence, (3) take the exam while the Signal and Vision collectors run silently in the background, and (4) submit — no video is ever uploaded. The instructor's flow is a single dashboard view.

Fig. 7 is a conceptual mockup of that Teacher Dashboard — not a live screenshot — illustrating the intended layout: a session list filterable by Live / Flagged / Calibration log / Fairness audit, with each row showing the student identifier, the current risk band, the single top-contributing signal from the explainability layer, the episode count, and a Review action. Consistent with Section III-F, no raw video is ever shown; only derived evidence and per-signal explanation chips reach the instructor.

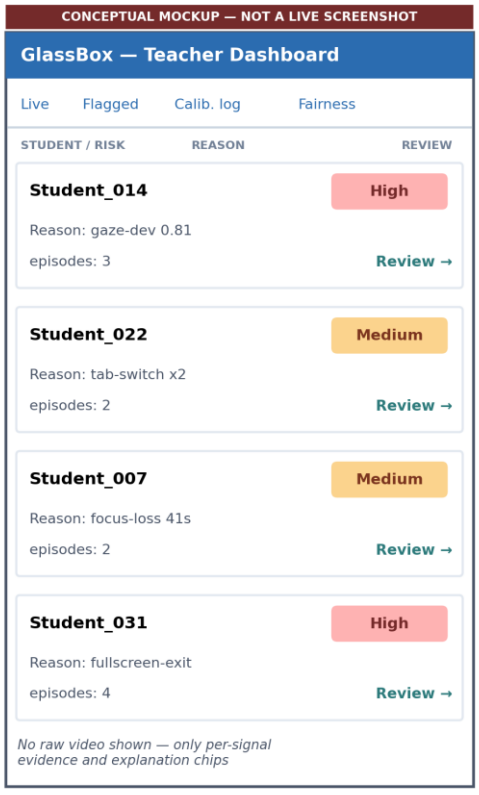


Fig. 7. Conceptual mockup of the GlassBox Teacher Dashboard — explicitly labeled as such to avoid implying an existing screenshot — organized around the Risk / Reason / Review flow a reviewer needs at a glance.

IX. LIMITATIONS

- A second device kept outside the camera frame cannot be detected by any browser-based system; this is a limitation of the approach in general, not unique to GlassBox.
- OS-level lockdown (blocking alt-tab or other native apps) is architecturally impossible for a browser-based tool and would require a native application with system hooks, as in Respondus LockDown Browser or Examplify.
- A sufficiently sophisticated student could attempt to game the calibration step beyond basic direction-validation checks.
- No raw video is available to instructors, a deliberate privacy trade-off that can make genuinely disputed cases harder to resolve; this is mitigated by an optional, disclosed, High-risk-only local clip.
- Evaluation currently rests on one external fairness dataset (FairFace) plus internal audits, not large-scale live deployment.
- The Logistic Regression model's linear decision boundary (Fig. 5b) may under-fit genuinely non-linear cheating patterns; this trade-off is accepted deliberately in exchange for interpretability, as argued in Section II.

X. CONCLUSION AND FUTURE WORK

GlassBox reframes exam-proctoring fairness as a calibration problem: rather than adding a new detection model or a separate rule for accommodated students, it scores every student against their own recorded baseline and shifts thresholds individually where needed, while keeping the scoring model interpretable and the raw biometric data off the server entirely. This extended report additionally makes the pipeline auditable end to end — Section III gives the exact formulas from raw signal to risk band, Section IV documents the internal structure of every model involved, and Section VII turns the positioning claim into a checkable table.

Current work has completed the zeroth review and an initial live implementation (React + Vite frontend with MediaPipe Tasks Vision integrated, and an initial test session recorded to SQLite), with dataset collection across the team beginning next. Future work includes completing the four-part evaluation in Section VI and replacing the illustrative templates in Fig. 6 with real k-fold results, pushing the project to a shared repository for parallel per-teammate data collection, running the five-reviewer trial described in Section VI, and finalizing the base-paper comparison in Section II with measured numbers rather than reported ones.

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